

G10S2 AP Calculus Review Practice

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Part A: Calculus

MCQ [30]

1. Evaluate the limit

$$\lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \right)^{1/x^2}.$$

- (a) $e^{-1/6}$
- (b) $e^{-1/3}$
- (c) $e^{1/6}$
- (d) $e^{1/3}$
- (e) 1

2. Compute the integral

$$\int_0^1 x e^{x^2} dx.$$

- (a) $\frac{e-1}{2}$
- (b) $e - 1$
- (c) $\frac{e+1}{2}$
- (d) $e + 1$
- (e) 1

3. Determine the convergence of the series

$$\sum_{n=1}^{\infty} \frac{n!}{n^n}.$$

- (a) Converges absolutely
- (b) Converges conditionally
- (c) Diverges

- (d) Test is inconclusive
- (e) Converges by the ratio test

4. Find the radius of convergence of the power series

$$\sum_{n=1}^{\infty} \frac{(x-2)^n}{n 3^n}.$$

- (a) 1
- (b) 3
- (c) e
- (d) ∞
- (e) $\frac{1}{3}$

5. Evaluate

$$\int_0^{\pi} x \sin x \, dx.$$

- (a) π
- (b) 2
- (c) 0
- (d) $\frac{\pi^2}{2}$
- (e) 1

6. The third-degree Taylor polynomial for $\ln(1+x)$ at $x=0$ is $T_3(x)$. Which of the following is a valid Lagrange-remainder bound for $R_3(0.5)$?

- (a) $\frac{1}{64}$
- (b) $\frac{1}{384}$
- (c) $\frac{(0.5)^3}{3!}$
- (d) $\frac{(0.5)^4}{4!}$
- (e) $\frac{(0.5)^4}{4}$

7. Find the general solution to the differential equation

$$\frac{dy}{dx} = \frac{y}{x} + x.$$

- (a) $y = Cx + x^2$
- (b) $y = Ce^x + x$

(c) $y = Cx^2 + x$

(d) $y = x^2 + C$

(e) $y = x^2 + Cx$

8. Compute the area enclosed by the polar curve

$$r = 2 \sin \theta.$$

(a) π

(b) 2

(c) 4

(d) $\frac{\pi}{2}$

(e) 2π

9. Determine the convergence of the alternating series

$$\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{\sqrt{n}}.$$

(a) Converges absolutely

(b) Converges conditionally

(c) Diverges

(d) Test is inconclusive

(e) Converges by the ratio test

10. Evaluate the improper integral

$$\int_0^{\infty} x e^{-x^2} dx.$$

(a) $\frac{1}{2}$

(b) 1

(c) 0

(d) ∞

(e) 2

11. For the parametric curve $x = t^2$, $y = t^3$, compute

$$\frac{d^2y}{dx^2} \quad \text{at } t = 1.$$

(a) $\frac{3}{4}$

(b) $\frac{3}{2}$

(c) $\frac{1}{2}$

- (d) $\frac{2}{3}$
- (e) $\frac{4}{3}$

12. Find the volume of the solid obtained by revolving $y = \sqrt{x}$, $0 \leq x \leq 4$, about the x -axis.

- (a) 8π
- (b) 4π
- (c) 16π
- (d) $\frac{32}{3}\pi$
- (e) 2π

13. Evaluate the limit

$$\lim_{x \rightarrow \infty} x(\sqrt{x^2 + 1} - x).$$

- (a) 1
- (b) $\frac{1}{2}$
- (c) 0
- (d) ∞
- (e) $-\frac{1}{2}$

14. Determine the convergence of the improper integral

$$\int_2^{\infty} \frac{1}{x(\ln x)^2} dx.$$

- (a) Converges (to $1/\ln 2$)
- (b) Diverges
- (c) Converges (to 1)
- (d) Converges (to ∞)
- (e) Converges (to $\frac{1}{2}$)

15. Compute

$$\int_0^{\pi/4} \tan^2 x \, dx.$$

- (a) $1 - \frac{\pi}{4}$
- (b) $\frac{\pi}{4}$
- (c) $\frac{\pi}{2} - 1$
- (d) 1
- (e) $\pi - 2$

FRQ [27]

Set 1: No Calculator

- Let $f(x) = xe^{-x}$.
 - Find the critical points of f and classify each as a local maximum, minimum, or neither.
 - Compute $\lim_{x \rightarrow \infty} x f(x)$.
- Evaluate the integral

$$\int_0^{\pi} x \cos x \, dx.$$

- Determine the radius and interval of convergence of the power series

$$\sum_{n=1}^{\infty} \frac{n(x-1)^n}{2^n}.$$

- Solve the differential equation

$$\frac{dy}{dx} = \frac{x+y}{x},$$

with the initial condition $y(1) = 0$.

Set 2: No Calculator

- Find the area of the region enclosed by $y = \sqrt{x}$ and $y = \frac{x}{2}$.
- Using the shell method, compute the volume of the solid obtained by revolving the region bounded by

$$y = x^2, \quad x = 0, \quad y = 4$$

about the y -axis.

- For the parametric curve

$$x = e^t \cos t, \quad y = e^t \sin t,$$

find

- $\frac{dy}{dx}$ at $t = 0$,
- $\int_{t=0}^{\ln 2} y \frac{dx}{dt} dt$.

- Determine whether the series

$$\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^2}$$

converges absolutely, converges conditionally, or diverges. Justify your answer.

Set 3: Calculator Allowed

1. Use one iteration of Newton's method with initial guess $x_0 = 0.5$ to approximate a solution to $\cos x = x$.
 - (a) Compute x_1 .
 - (b) Evaluate $|\cos x_1 - x_1|$.
2. Use a calculator to approximate

$$\int_1^3 e^{-x^2} dx$$

to four decimal places.

3. Using a Maclaurin series, find the first three nonzero terms of $\ln(1+x)$ and then use them to approximate $\ln(1.5)$.
4. Employ a graphing utility to find the x -coordinate of the intersection of

$$y = \ln x \quad \text{and} \quad y = x - 2$$

accurate to two decimal places.

Part B: Precalculus [43]

1. Graph: $\tan x + \sec x$. [3]
2. If $m > 1$ satisfies $\log_9(\log_8(m)) = 2024$, then $\log_3(\log_2(m))$ is? [3]
3. Solve each of the following trigonometric equations in $[0, 2\pi]$: [2*2]
 - (a) $\sin(3x) - \sin x = \sqrt{3} \cos(2x)$.
 - (b) $1 - \cos x = 2 \sin^2 x$.
4. The price of Up-N-Down Stock varies sinusoidally with time. At the start of this year, the price of the stock is \$70 per share. Over the next 4 days, the price climbs to a peak of \$80 per share. If the lowest price the stock reaches is \$40 per share, how much money can you make this year (in 365 days starting from the start of this year) if you start with \$700 and time all of your trades perfectly? (Assume that you can buy and sell stock as many times as you want, on any day that you want including weekends, but you can never sell stock that you haven't already bought first). [4]
5. Simplify, for $x \in [2\pi, 3\pi]$: [2*3]
 - (a) $\arccos(\sin x)$
 - (b) $\tan(\arccos(x))$
 - (c) $\cos(\arcsin x)$

6. In triangle $\triangle ABC$, we have $\cos C = \frac{\sin A + \cos A}{2} = \frac{\sin B + \cos B}{2}$. Find the value of $\cos C$. [8]
7. In the Cartesian plane, let hyperbola $\Gamma : x^2 - y^2 = 1$ have right vertex A at $(1, 0)$. Let all circles with exactly one intersection with each branch of Γ be "cool". Two cool circles are externally tangent at P with distance d between their centers. Find all possible $\frac{d}{|PA|}$. The following steps are to guide you and each carries credit:
- Articulate what property the center of cool circles have to satisfy. A formal proof is not needed. [1]
 - Let the center of the circle be $(0, y_0)$, write the equation of the circle. Find a relationship between y_0 and r_0 based on the discriminant. [4]
 - Let the two cool circles in question have centers $(0, h + r_1)$ and $(0, h - r_2)$, plug both these values into the result from the previous step to obtain h in terms of r_1 and r_2 . [4]
 - Express everything in terms of r_1 and r_2 , using the results from the previous step to simplify to a final answer to the problem. [6]